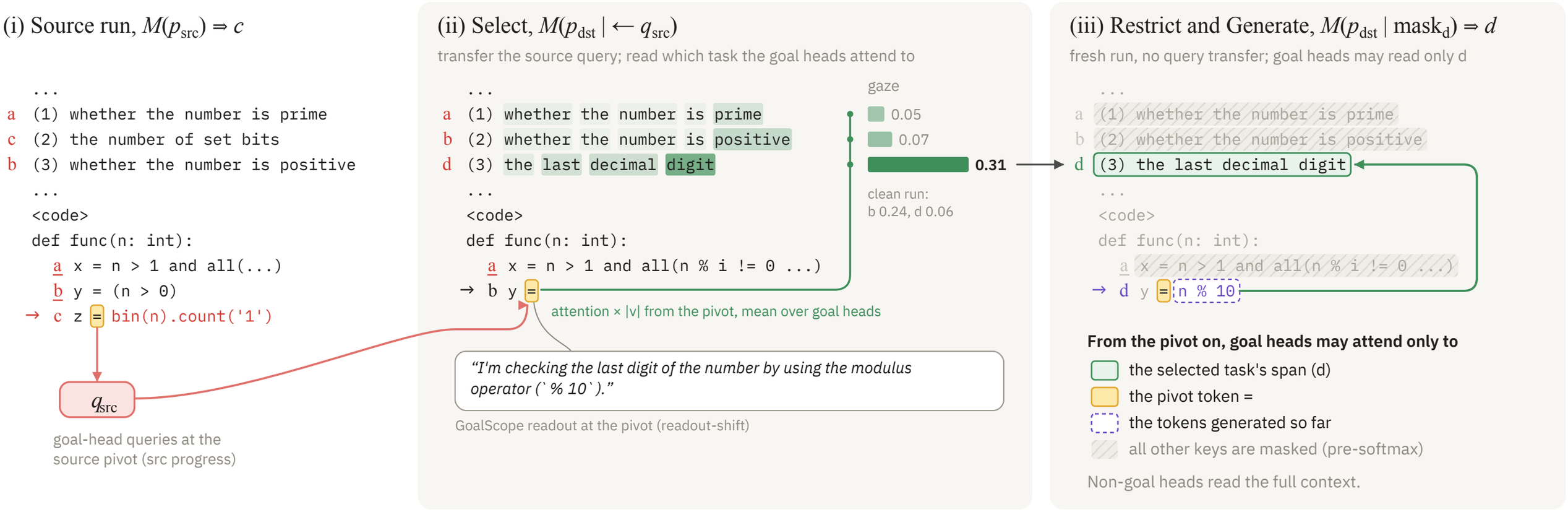

(i) Source run, $M(p_{\text{src}}) \Rightarrow c$
...
a (1) whether the number is prime
c (2) the number of set bits
b (3) whether the number is positive
...
<code>
def func(n: int):
a x = n > 1 and all(...)
b y = (n > 0)
→ c z = bin(n).count('1')
q_{src}
goal-head queries at the source pivot (src progress)
(ii) Select, $M(p_{\text{dst}} \mid \leftarrow q_{\text{src}})$
transfer the source query; read which task the goal heads attend to
...
a (1) whether the number is prime
b (2) whether the number is positive
d (3) the last decimal digit
...
<code>
def func(n: int):
a x = n > 1 and all(n % i != 0 ...)
→ b y =
gaze
0.05
0.07
0.31
clean run:
b 0.24, d 0.06
attention × |v| from the pivot, mean over goal heads
"I'm checking the last digit of the number by using the modulus operator (` % 10 `)."
GoalScope readout at the pivot (readout-shift)
(iii) Restrict and Generate, $M(p_{\text{dst}} \mid \text{mask}_{\text{d}}) \Rightarrow d$
fresh run, no query transfer; goal heads may read only d
...
a (1) whether the number is prime
b (2) whether the number is positive
d (3) the last decimal digit
...
<code>
def func(n: int):
a x = n > 1 and all(n % i != 0 ...)
→ d y = n % 10
From the pivot on, goal heads may attend only to
the selected task's span (d)
the pivot token =
the tokens generated so far
all other keys are masked (pre-softmax)
Non-goal heads read the full context.